

Practical 5.1 - 0/1 Knapsack Problem

```
#include <iostream>

using namespace std;

int max(int a, int b) {
    return (a > b) ? a : b;
}

int main() {
    int n, W;

    cout << "Enter number of items: ";

    cin >> n;

    int wt[n], val[n];

    for(int i = 0; i < n; i++) {
        cout << "Enter weight and profit of item " << i + 1 << ": ";

        cin >> wt[i] >> val[i];
    }

    cout << "Enter knapsack capacity: ";

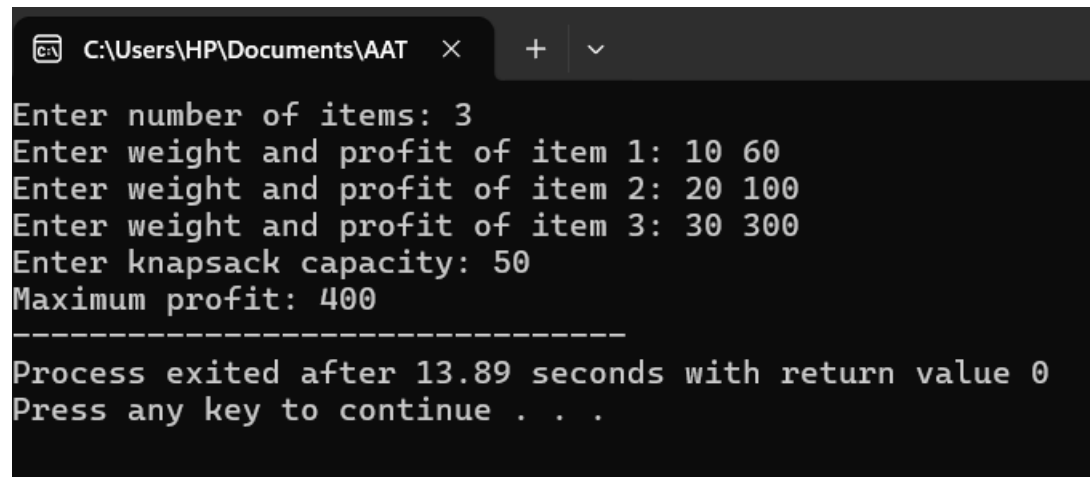
    cin >> W;

    int dp[n + 1][W + 1];

    for(int i = 0; i <= n; i++) {
        for(int w = 0; w <= W; w++) {
            if(i == 0 || w == 0)
                dp[i][w] = 0;
            else if(wt[i - 1] <= w)
                dp[i][w] = max(val[i - 1] + dp[i - 1][w - wt[i - 1]], dp[i - 1][w]);
            else
                dp[i][w] = dp[i - 1][w];
        }
    }

    cout << "Maximum profit: " << dp[n][W];
```

```
return 0;  
}
```



The screenshot shows a Windows command prompt window with the title bar "C:\Users\HP\Documents\AAT". The window contains the following text:

```
Enter number of items: 3  
Enter weight and profit of item 1: 10 60  
Enter weight and profit of item 2: 20 100  
Enter weight and profit of item 3: 30 300  
Enter knapsack capacity: 50  
Maximum profit: 400  
-----  
Process exited after 13.89 seconds with return value 0  
Press any key to continue . . .
```

5.2 - Coin Change-Making Problem

```
#include <iostream>

#include <climits>

using namespace std;

int min(int a, int b) {
    return (a < b) ? a : b;
}

int main() {
    int n, amount;

    cout << "Enter number of coin types: ";
    cin >> n;

    int coins[n];
    cout << "Enter coin denominations:\n";
    for(int i = 0; i < n; i++)
        cin >> coins[i];

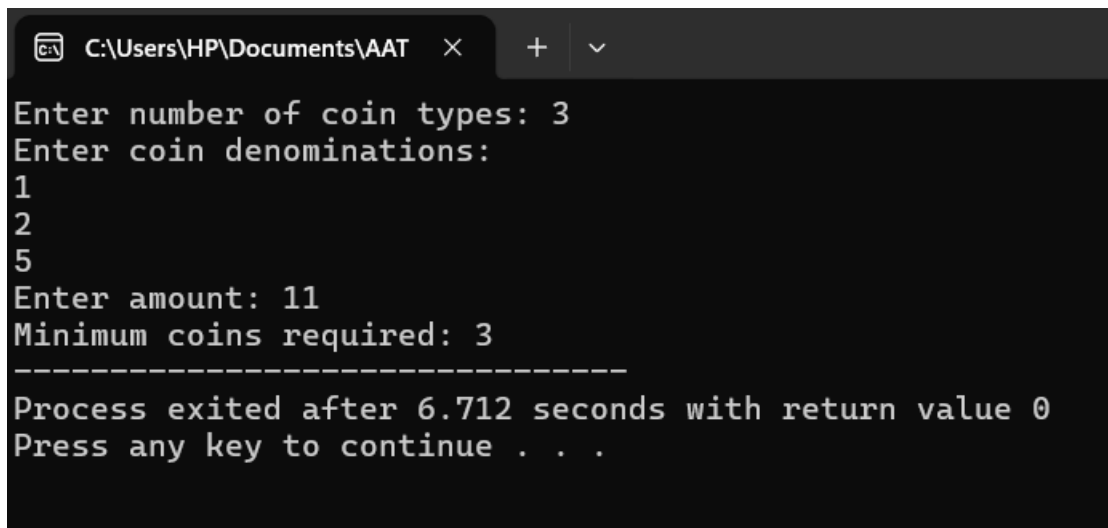
    cout << "Enter amount: ";
    cin >> amount;

    int dp[amount + 1];
    dp[0] = 0;

    for(int i = 1; i <= amount; i++)
        dp[i] = INT_MAX;

    for(int i = 1; i <= amount; i++) {
```

```
    for(int j = 0; j < n; j++) {  
        if(coins[j] <= i && dp[i - coins[j]] != INT_MAX)  
            dp[i] = min(dp[i], dp[i - coins[j]] + 1);  
    }  
}  
  
cout << "Minimum coins required: " << dp[amount];  
  
return 0;  
}
```



The screenshot shows a Windows command prompt window with the title bar "C:\Users\HP\Documents\AAT". The program prompts the user for the number of coin types (3), the coin denominations (1, 2, 5), and the amount (11). It then outputs "Minimum coins required: 3". A separator line of dashes follows. The final output states "Process exited after 6.712 seconds with return value 0" and "Press any key to continue . . .".

```
C:\Users\HP\Documents\AAT > Enter number of coin types: 3  
Enter coin denominations:  
1  
2  
5  
Enter amount: 11  
Minimum coins required: 3  
-----  
Process exited after 6.712 seconds with return value 0  
Press any key to continue . . .
```

5.3 - Weighted Job Scheduling Problem

```
#include <iostream>

#include <algorithm>

using namespace std;

struct Job {
    int start, finish, profit;
};

bool compare(Job a, Job b) {
    return a.finish < b.finish;
}

int main() {
    int n;
    cout << "Enter number of jobs: ";
    cin >> n;

    Job jobs[n];

    for(int i = 0; i < n; i++) {
        cout << "Enter start, finish and profit: ";
        cin >> jobs[i].start >> jobs[i].finish >> jobs[i].profit;
    }

    sort(jobs, jobs + n, compare);

    int dp[n];
    dp[0] = jobs[0].profit;
```

```

for(int i = 1; i < n; i++) {
    int incl = jobs[i].profit;

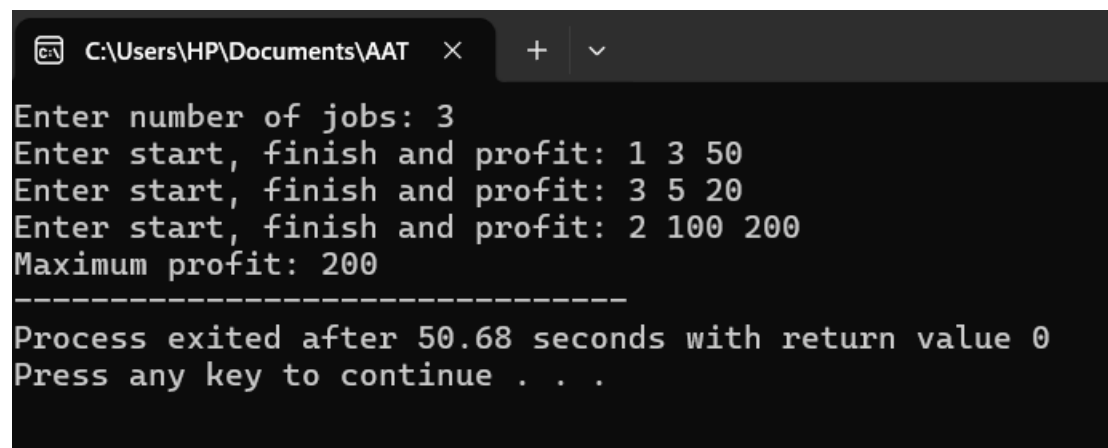
    for(int j = i - 1; j >= 0; j--) {
        if(jobs[j].finish <= jobs[i].start) {
            incl += dp[j];
            break;
        }
    }

    dp[i] = max(incl, dp[i - 1]);
}

cout << "Maximum profit: " << dp[n - 1];

return 0;
}

```



```

C:\Users\HP\Documents\AAT
Enter number of jobs: 3
Enter start, finish and profit: 1 3 50
Enter start, finish and profit: 3 5 20
Enter start, finish and profit: 2 100 200
Maximum profit: 200
-----
Process exited after 50.68 seconds with return value 0
Press any key to continue . . .

```